UNIVERSITY OF HELSINKI

NETWORK ANALYSIS

HOMEWORK

EXERCISE 1

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PROBLEM 1

1.1

For the graph **air_traffic**: Number of Nodes = **1226**, Number of Edges = **2410**

For the graph **academic_collaboration**: Number of Nodes = **5242**, Number of Edges = **14496**

1.2

(i)

Fig.1 and Fig.2 show the distribution of **air_traffic**, **academic_collaboration**, $G(n, m)$ and $B(n, m)$.
From the figures, we can see that **real word** graph (**air_traffic** and **academic_collaboration**) are similar to $B(n, m)$. All distributions are skewed to the left.

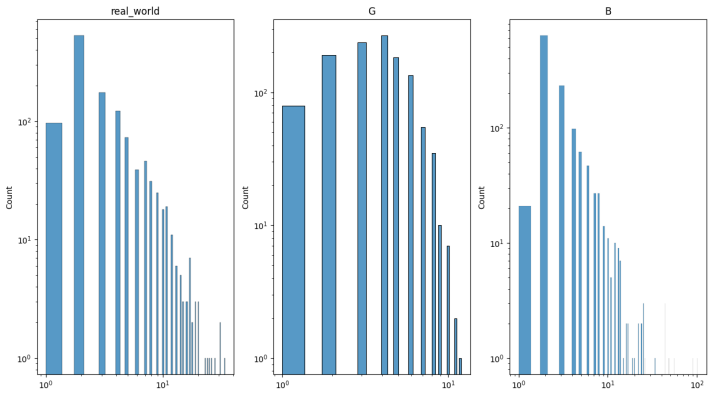

Figure 1: Node distributions of graph **air_traffic**, $G(n, m)$ and $B(n, m)$.

(ii)

Clustering coefficient of **air_traffic** = 0.0675
Clustering coefficient of $G(1226, 2410)$ = 0.0034
Clustering coefficient of $B(1226, 2410)$ = 0.0279

Clustering coefficient of **academic_collaboration** = 0.5296
Clustering coefficient of $G(5242, 14496)$ = 0.0006
Clustering coefficient of $B(5242, 14496)$ = 0.0075

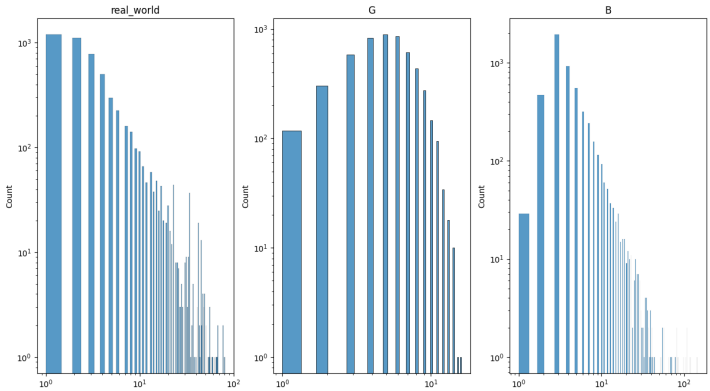

Figure 2: Node distributions of graph **academic_collaboration**, $G(n, m)$ and $B(n, m)$.

We can see that with the case **air_traffic**, the real world graph' clustering coef. is more similar to $B(n, m)$ whereas for the case **academic_collaboration**, the clustering coef. of real world graph, $G(n, m)$ and $B(n, m)$ are very different.

Problem 2

Q: *1. What is the expected degree of any given vertex v?*

A: The expected degree is: $\mathbf{E}[d] = p \times (n-1) = c$

Q: *2. Let $v \in V$ be one of the nodes. What is the probability of that v is isolated?*

A: The probability that v is isolated is $(1-p)^{n-1} = \left(1 - \frac{c}{n-1}\right)^{n-1}$

Fig.3 illustrates different c values in range $[0, 10]$ and probability of node being isolated.

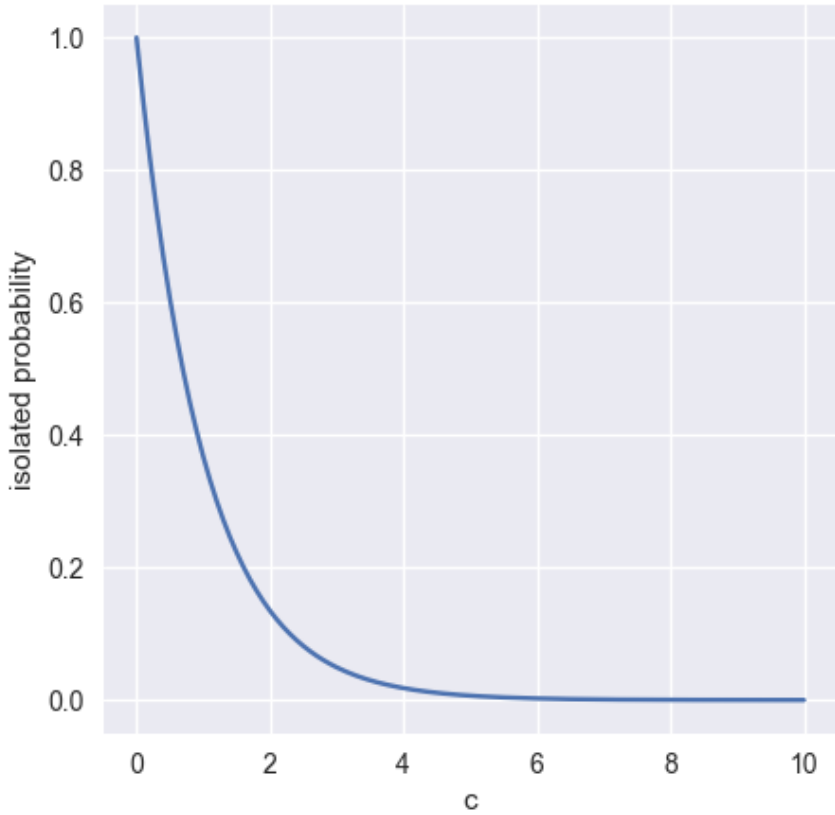

Figure 3: Different c values and probability of node being isolated.

Denote the probability of given node v being isolated as p_{iso}.

We can see that, when $c = 0$, $p_{iso} = 1$ - which means the graph is completely isolated. However, when c increases, p_{iso} decreases exponentially quickly and is asymptotic to 0.

2.1 Connectivity

Although the problem only requires to report the single smallest c value with which the $G(n,p)$ is connected, I do further. In particular, I run the experiment many times to find the smallest satisfying c values (Fig.4) and I recognize that:

The smallest satisfying c value is not fixed.

The reason is that the $G(n,p)$ is a probabilistic model, everything is not fixed.

I also calculate the corresponding p values for each c and see that the p is below the threshold $\frac{\log n}{n}$, which is the informal boundary for the p so that the graph $G(n,p)$ is connected. However, It is understandable because that boundary is not the exact boundary (Fig.5).

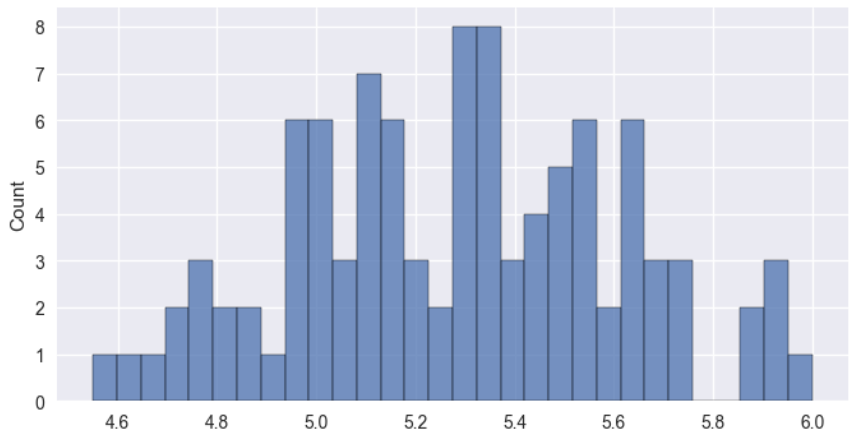

Figure 4: Histogram of the smallest c values with which $G(n, p)$ is connected.

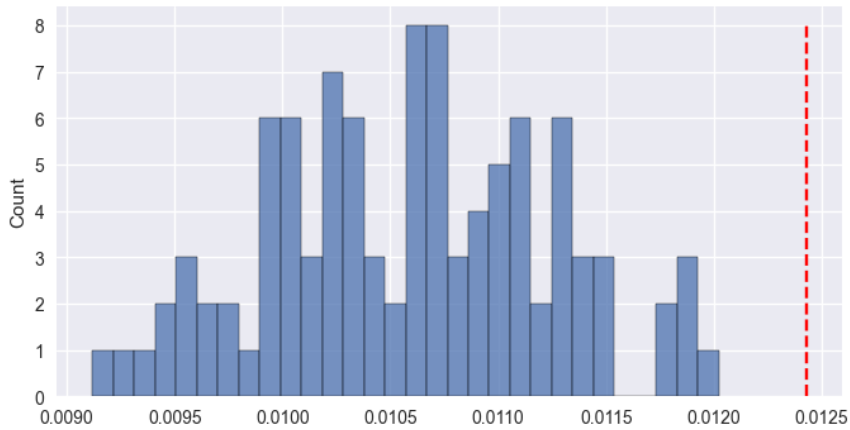

Figure 5: Histogram of the smallest p values with which $G(n, p)$ is connected. The red dashed line indicates the informal boundary $\frac{\log n}{n}$ with $n = 500$.